

DAYANANDA SAGAR ACADEMY OF TECHNOLOGY AND MANAGEMENT

Autonomous under VTU, Belagavi

Accredited by NAAC with Grade A+



Project Based Learning- PCB Design (BEC306)

Project Report

on

“DTMF Based Industry Motor Controller ”

Submitted by

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In partial fulfillment of the requirement for the degree of

BACHELOR OF ENGINEERING

In

ELECTRONICS & COMMUNICATION ENGINEERING

Under the Guidance of

Dr Saranya S N

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Department of Electronics and Communication Engineering

Accredited by NBA, New Delhi.

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Udayapura, Kanakapura Road, Bengaluru-560082

2024-2025

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Department of Electronics and Communication Engineering



This is to satisfy that the **Project Based Learning- PBC Design** project work entitled “**DTMF Based Industry Motor Controler**” carried out by **Tejas Adithya S B (1DT23EC106)**, **Vikyath J (1DT23EC121)**, **Yashas B(1DT23EC124)**, **Yashwanth S (1DT23EC125)**, a bonafide student of **DAYANANDA SAGAR ACADEMY OF TECHNOLOGY AND MANAGEMENT** in Bachelor of Engineering in Electronics and Communication Engineering of Visvesvaraya Technological University, Belagavi during the academic year 2024-2025. It is certified that all the corrections/suggestions indicated for internal assessment have been incorporated in the report. The project report has been approved as it satisfies the academic requirement in respect of the Project Based Learning course prescribed for the degree.

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His/Her Performance during Course was excellent in the domain of **PCB Design and Development for Industry Product**. He/She during the course demonstrated the critical thinking and learnability towards the problem solving in product development for various applications.

Mr. Kotresh Mundrugi
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Yours Sincerely

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ABSTRACT

This report presents the design and implementation of a Dual-Tone Multi-Frequency (DTMF)-based industrial motor controller, showcasing an innovative approach to remotely operating and monitoring motors using a mobile phone. The system leverages the DTMF signaling method, wherein each button press on a mobile keypad generates a unique frequency pair that serves as a control signal. These signals are decoded and processed to manage motor operations such as start, stop, and direction control.

The project integrates key components, including a DTMF decoder, microcontroller, relay driver circuitry, and the motor, ensuring seamless communication and precise execution of commands. Our contributions include the design of a highly efficient DTMF decoding system to enhance signal interpretation accuracy, the development of a robust microcontroller program for error-free motor control, and optimization of the relay driver circuit for reliable operation under industrial conditions.

Additionally, we addressed challenges such as noise interference in DTMF signal reception and ensured the safety and reliability of the system by implementing protective measures against short circuits and signal distortion. The designed system offers significant advantages, including low-cost implementation, ease of operation, and real-time control from any location, making it an ideal solution for industries requiring remote motor management.

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Chapter 1

Problem Definition

Industrial automation plays a pivotal role in enhancing productivity, reliability, and safety across various sectors. In manufacturing plants, refineries, and other industrial settings, motors are key components driving numerous operations. Traditional methods of motor control often involve manual intervention or require complex wired systems for operation, which are susceptible to limitations such as high installation costs, limited mobility, and vulnerability to wear and tear. As industries strive for more efficient, cost-effective, and remotely manageable solutions, the need for innovative motor control systems becomes increasingly apparent.

The primary problem addressed in this project is the design and development of a system capable of controlling industrial motors remotely, leveraging modern communication technologies. Current solutions, such as programmable logic controllers (PLCs) and SCADA systems, while effective, often entail substantial investment and may not be practical for small or medium-sized enterprises. Furthermore, manual control systems limit flexibility, and conventional wireless control systems may suffer from issues like interference, high power consumption, or range constraints.

To overcome these challenges, the proposed solution is to design a chip-based motor controller using Dual-Tone Multi-Frequency (DTMF) technology. DTMF, a signaling method widely used in telecommunication systems, offers a simple yet robust mechanism for transmitting commands remotely. By utilizing the DTMF signals generated from a mobile device, the motor controller can process user inputs to perform various motor operations such as starting, stopping, and direction control. This approach combines the advantages of low-cost implementation, ease of operation, and wide accessibility, addressing the limitations of existing systems.

Key Challenges in Existing Systems

1. **Wired Control Systems:** These systems often require extensive wiring, leading to high installation and maintenance costs. Additionally, they lack flexibility and are difficult to

adapt to changing industrial layouts.

2. **Cost of Automation:** Advanced industrial automation systems, such as PLCs, involve significant initial investment, making them less feasible for small-scale industries.
3. **Limited Remote Control Options:** Existing remote-control technologies may face challenges such as limited range, susceptibility to signal interference, and reliance on proprietary hardware.
4. **Manual Operation:** Manual control systems necessitate on-site presence, which can lead to delays, increased labor costs, and reduced operational efficiency.
5. **Reliability Issues:** Traditional control systems may not be robust against external factors like electrical noise, signal distortion, or environmental conditions, leading to potential operational failures.

Problem Scope and Requirements

To design a functional DTMF-based motor controller, the system must meet the following technical and operational requirements:

1. **DTMF Signal Decoding:** Develop a decoding mechanism to accurately interpret the DTMF tones generated by a mobile phone and translate them into actionable commands.
2. **Microcontroller Programming:** Implement efficient firmware on a microcontroller to process decoded signals and execute the corresponding motor control actions.
3. **User Interface:** Ensure an intuitive and user-friendly interface for operators, simplifying system operation and minimizing the learning curve.
4. **Hardware Design:** Create a compact, durable, and cost-effective hardware solution, suitable for industrial environments.
5. **System Integration:** Seamlessly integrate all components, ensuring smooth communication between the DTMF decoder, microcontroller, and motor driver circuits.

Chapter 2

Objectives

- 2.1** The proposed DTMF-based motor controller aims to address the limitations of existing systems by offering a cost-effective, reliable, and user-friendly solution for remote motor operation. The specific objectives include:
1. **Remote Operation:** Enable operators to control industrial motors from any location using a standard mobile phone.
 2. **Cost Efficiency:** Develop a low-cost solution that leverages readily available DTMF technology, making it accessible to small and medium-sized enterprises.
 3. **Simplicity and Reliability:** Design a system with straightforward operation and high reliability under industrial conditions.
 4. **Scalability:** Create a flexible architecture that can be easily scaled to accommodate multiple motors or additional functionalities.
 5. **Safety:** Incorporate safety features to protect the system and connected motors from short circuits, signal distortion, and other potential hazards.
 6. **Provide Real-Time Feedback:** Integrate mechanisms to deliver real-time status updates to the operator, ensuring that motor operations are monitored and issues are promptly addressed
 7. **Facilitate Easy Maintenance:** Design the system with modular components and diagnostic features, simplifying maintenance and reducing downtime in case of faults.
 8. **Promote Scalability and Adaptability:** Create a flexible system that can be upgraded or customized to suit different industrial needs, ensuring long-term usability and value.

Chapter 3

Methodology

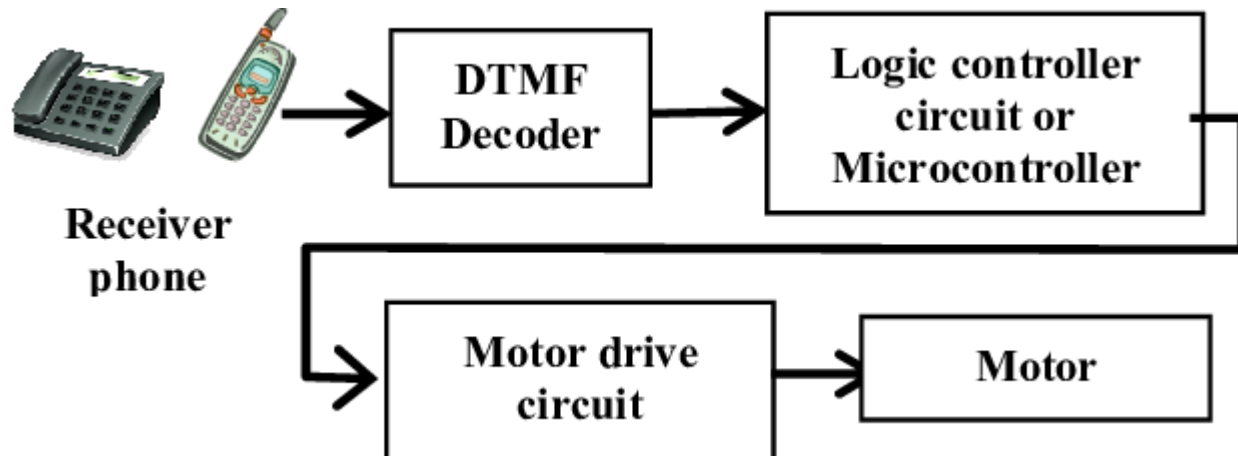


Fig 3.1 FlowChart

3.1 Understanding Connections

3.1.1 System Requirements Analysis

- **Define Goals:** Understand the basic functions needed, like starting, stopping, and controlling the motor remotely.
- **List Components:** Identify key components, such as a DTMF decoder, a microcontroller, LEDs for indicators, and basic circuitry to control operations.
- **Environmental Considerations:** Account for factors like power needs and potential interference in the industrial setting.

3.1.2 Component Selection

- **Signal Decoder:** Choose a simple decoder (e.g., CM8870) to convert DTMF signals into usable commands.
- **Control Unit:** Pick a microcontroller that can interpret signals and execute basic tasks.
- **LED Indicators:** Use LEDs to visually indicate the status of the system.
- **Resistors:** Include resistors with ratings of 1k Ω , 100k Ω , and 200k Ω for proper current limiting and circuit stability.
- **Capacitors:** Use capacitors with a rating of 0.1 μ F for signal coupling and filtering.
- **Power Supply:** Ensure adequate power to run all parts of the system reliably.

3.1.3 Circuit Design

- **Signal Flow:**
 - Connect a phone's output to the decoder for tone recognition.
 - Use a 0.1 μ F capacitor to couple the audio signal to the input of the CM8870, ensuring signal clarity.
- **Control Connections:**
 - Link the decoder's outputs to the microcontroller for signal interpretation.
 - Drive LEDs directly through appropriate resistors to indicate system states.
- **Simplification:** Keep the design as straightforward as possible to ensure easy assembly and troubleshooting.

3.1.4 Software Development

- **Basic Logic:**
 - Create simple programs to interpret the commands from the decoder and perform tasks like turning the motor on or off (or equivalent operations).
- **Error Handling:**
 - Include basic checks to handle incorrect or unclear signals.
- **Indicators:**
 - Add simple feedback mechanisms using LEDs to indicate system status, such as power on/off or signal detected.

3.1.5 Prototyping and Testing

- **Build a Prototype:**
 - Assemble all parts on a breadboard for initial testing.
 - Confirm that the circuit works as intended under normal conditions.
- **Basic Testing:**
 - Use a mobile phone to send commands and verify the LEDs respond appropriately to indicate system functionality.
 - Address any issues that arise, such as unclear signals or delays.

3.1.6 Final Adjustments

- **Refine the Design:**
 - Make the circuit more compact and robust for regular industrial use.
- **Durability:**
 - Opt for components that can handle variations in industrial environments.

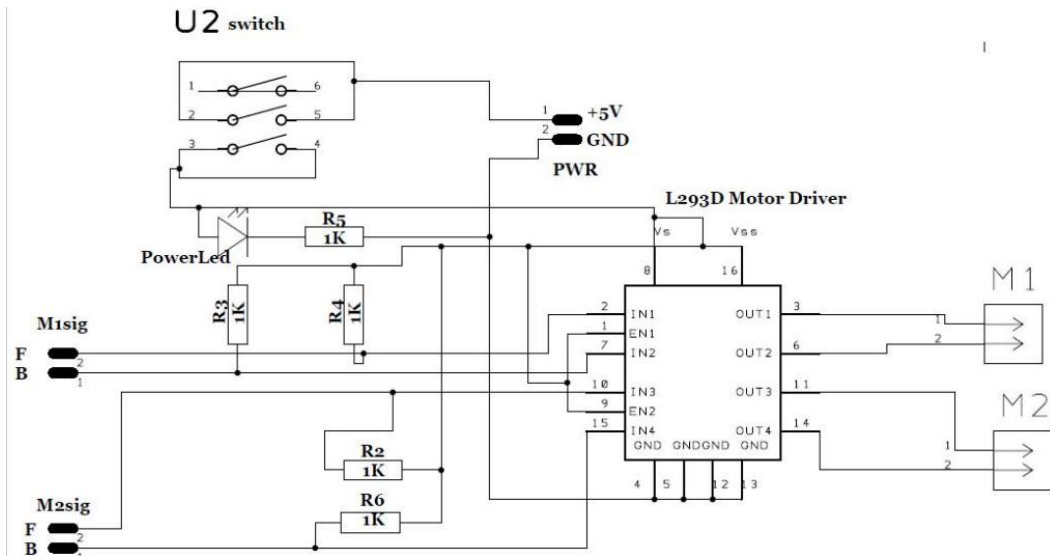


Fig 3.2 Connections of L293D Motor Driver

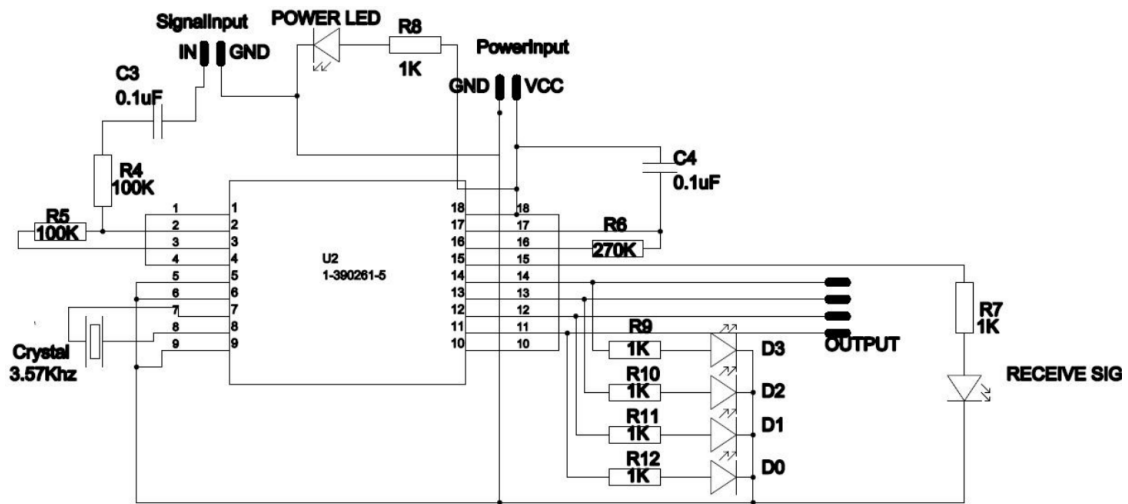


Fig 3.3 Connections of 1-390261 DTMF Decoder

3.2 Understanding flowchart:

3.2.1. DTMF Input

The system begins with the user generating DTMF signals via a telephone or mobile phone. Each button press on the phone produces a unique dual-tone frequency combination. These signals are sent through a wired or wireless connection to the system.

- Purpose: The phone acts as the remote input device, allowing commands to be sent from a distance.
 - Key Role: The DTMF tones serve as the primary communication method between the user and the controller.
-

3.2.2. DTMF Decoder

The DTMF signals are received by the DTMF Decoder (e.g., CM8870). This block is responsible for converting the audio frequency signals into corresponding digital binary outputs.

- Process: The decoder matches the received frequency pair to a predefined digital code.
 - Output: A 4-bit binary code is generated, which represents the specific key pressed by the user.
-

3.2.3. Logic Controller Circuit or Microcontroller

The binary output from the decoder is fed into the Logic Controller Circuit or Microcontroller. This is the central processing unit of the system and plays a crucial role in decision-making.

- Functions:
 - Interpret the received binary code.
 - Map the binary code to specific motor operations such as start, stop, or change direction.
 - Implement safety mechanisms like rejecting invalid commands or handling signal interference.
 - Flexibility: A microcontroller can also be programmed to perform additional logic, such as timing or prioritizing certain operations.
-

3.2.4. Motor Drive Circuit

The control signals from the microcontroller are sent to the Motor Drive Circuit. This block is responsible for driving the motor based on the received commands.

- Components:
 - Power transistors or motor driver ICs to handle the current required by the motor.
 - Protection components like diodes to safeguard against back-EMF from the motor.
 - Purpose: Convert the low-power control signals from the microcontroller into high-power signals suitable for motor operation.
-

3.2.5. Motor

The final block in the system is the Motor, which executes the desired operations as commanded by the user.

- Types of Operations:
 - Start or stop the motor.
 - Change the motor's rotation direction.
 - Adjust speed if the system supports variable control.
 - Outcome: The motor performs its intended task, such as operating an industrial machine, conveyor belt, or other mechanical systems.
-

Flow of Control

1. The user inputs a command using a phone.
2. The DTMF decoder converts the audio signal to a binary code.
3. The microcontroller processes the binary code and determines the required motor operation.
4. The motor driver circuit amplifies the control signals to drive the motor.
5. The motor executes the operation, and the process repeats as needed.

This system provides a simple, cost-effective, and remote-controlled solution for industrial motor automation.

Chapter 4

Components used and their descriptions

4.1 DTMF:

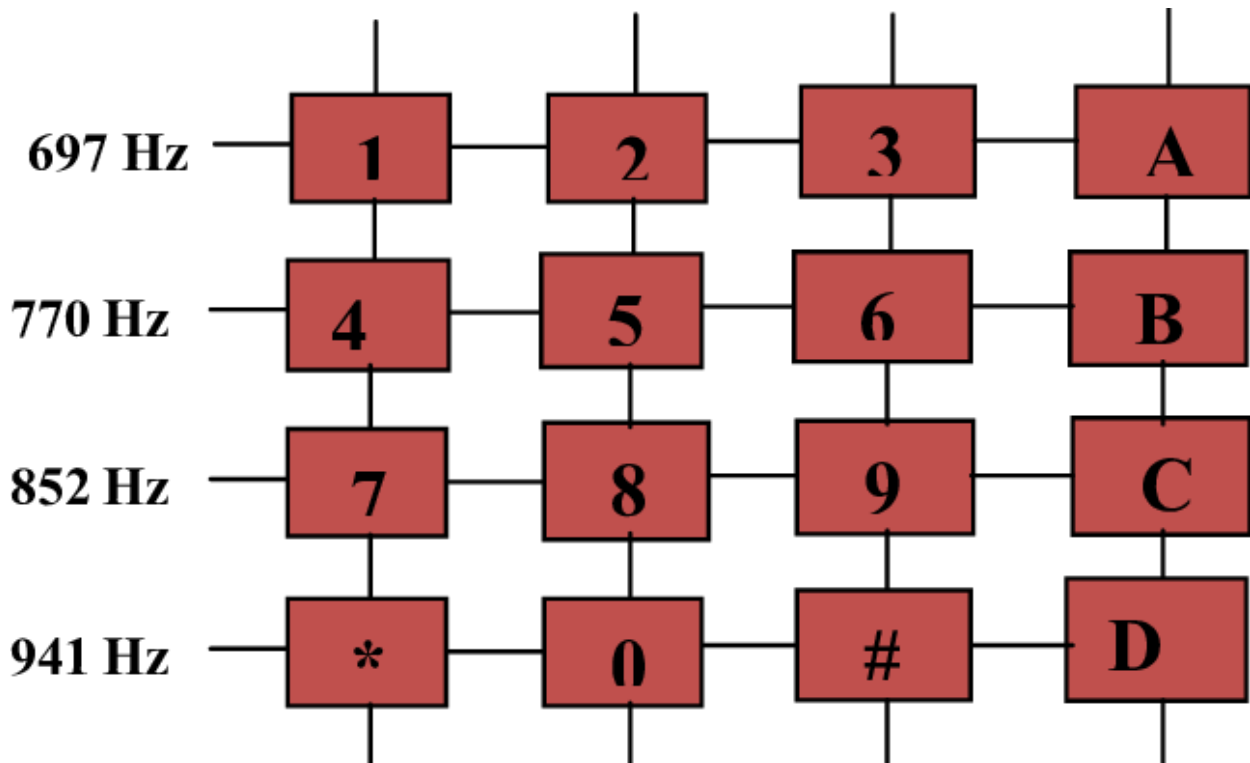


Fig 4.1 Signals in dtmf

Dual-Tone Multi-Frequency signaling (DTMF) is an in-band telecommunications signaling system using the voice-frequency band over telephone lines between telephone equipment and other communications devices and switching centers. The working procedure of DTMF is like dialing a customer care number followed by automatic voice recorded instructions. It will ask to press 1, 2 or any other number, when you press number in your mobile one particular procedure is happening, this is called a DTMF technology. If any button is pressed on the mobile phone keypad, then it will generate two frequencies. These tones are called row or column frequencies. Normally row frequencies are low frequencies and column frequencies are high frequencies. DTMF keypad is placed out on a 4 cross 4 matrices, in which each row represents low frequency, each column represents high frequency, with DTMF, each key pressed on a phone generates two tones of the specific frequencies one tone is generated from a high frequency tones and low frequency tone. These tones are converted to digital form using DTMF decoder circuit. These codes are the address of the destination which is read and preceded by the computer that connects the caller to the destination.

4.1.1 Pin Diagram of CM8870 DTMF Decoder

The CM8870 is a widely used DTMF decoder IC that converts audio tones into digital signals. Its pin configuration is simple and essential for implementing DTMF-based systems. Below is a detailed explanation of each pin:

4.1.1.1. Power Pins

- Pin 1 (VDD): The positive power supply pin. Typically connected to a +5V DC source.
 - Pin 2 (GND): Ground pin. Connect this to the negative terminal of the power supply to complete the circuit.
-

4.1.1.2. Input Pins

- Pin 3 (IN+): The non-inverting input for the DTMF signal. Connect this pin to the audio signal source (e.g., phone output) through a capacitor for coupling.
 - Pin 4 (IN-): The inverting input. It is generally grounded or connected to a biasing circuit.
-

4.1.1.3. Oscillator Pins

- Pin 7 (OSC1) and Pin 8 (OSC2): These pins are used to connect an external crystal oscillator, typically 3.579 MHz. The crystal ensures accurate frequency decoding.
-

4.1.1.4. Decoded Output Pins

- Pins 11, 12, 13, 14 (Q1, Q2, Q3, Q4): These are the digital outputs that represent the decoded DTMF signal. Each combination of high (1) and low (0) levels corresponds to a specific keypress on a DTMF keypad.
-

4.1.1.5. Control and Status Pins

- Pin 5 (GS): This pin outputs the gain setting for the input amplifier. It helps in adjusting the input signal level.
- Pin 15 (STD): Stands for 'Signal Detect'. This pin goes high when a valid DTMF tone pair is detected.
- Pin 10 (ESt): End of Signal flag. It indicates the end of a DTMF signal.

4.1.1.6. Filter Pins

- Pin 6 (VRef): Provides a reference voltage for the internal filter circuitry.
- Pins 9 and 16 (FILT and TOE): Used for external filtering and control purposes to enhance signal clarity.

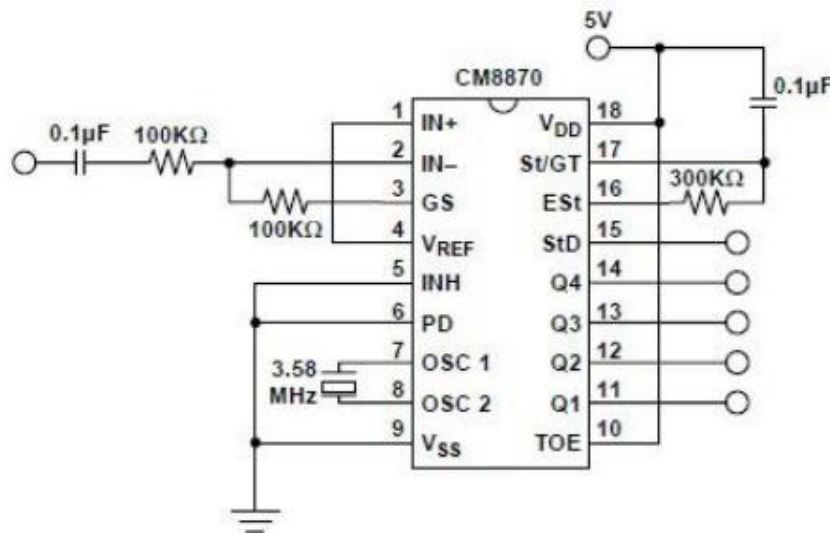


Fig 4.2 Pin diagram of DTMF Decoder

4.2 L293D IC

4.2.1 Description

L293D is a typical Motor driver or Motor Driver IC which allows DC motor to drive in either direction. L293D is a 16-pin IC which can control a set of two DC motors simultaneously in any direction. It means that you can control two DC motor with a single L293D IC.

4.2.2 Concept

It works on the concept of H-bridge. H-bridge is a circuit which allows the voltage to be flown in either direction. As you know voltage needs to change its direction for being able to rotate the motor in clockwise or anticlockwise direction, Hence H-bridge IC are ideal for driving a DC motor. In a single L293D chip there are two H-bridge circuit inside the IC which can rotate two DC motors independently. Due to its size it is very much used in robotic application for controlling DC motors. Given below is the pin diagram of a L293D motor controller. There are two Enable pins on L293D. Pin 1 and pin 9, for being able to drive the motor, the pin 1 and 9 need to be high. For driving the motor with left H-bridge you need to enable pin 1 to high. And for right H-Bridge you need to make the pin 9 to high. If any one of the either pin 1 or pin 9 goes low then the motor in the corresponding section will suspend working. It's like a switch

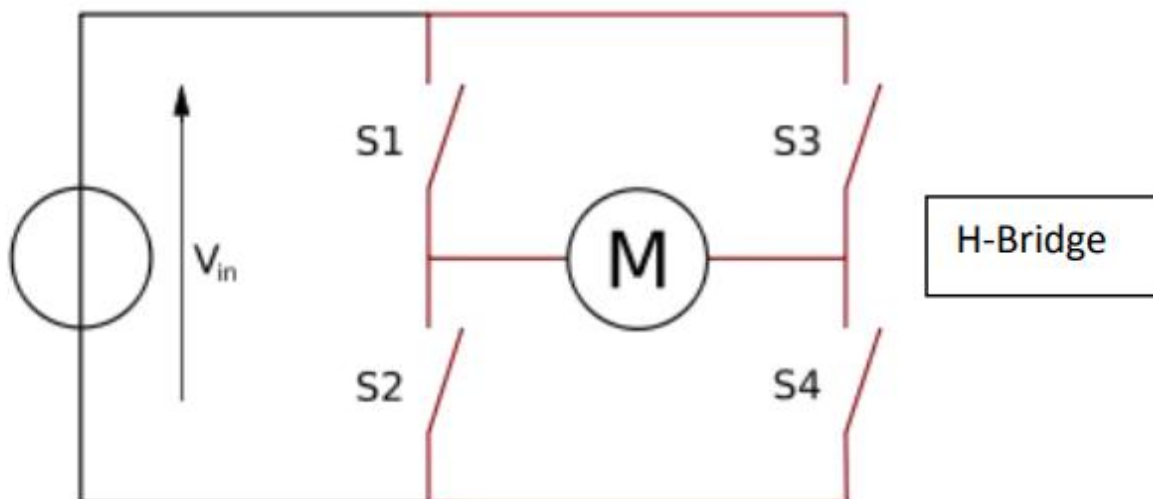


Fig 4.3 Working of L293D Motor driver

4.2.3 Working of L293D

There are 4 input pins for L293D, pin 2,7 on the left and pin 15,10 on the right as shown on the pin diagram. Left input pins will regulate the rotation of motor connected across left side and right input for motor on the right hand side. The motors are rotated on the basis of the inputs provided across the input pins as LOGIC 0 or LOGIC 1.

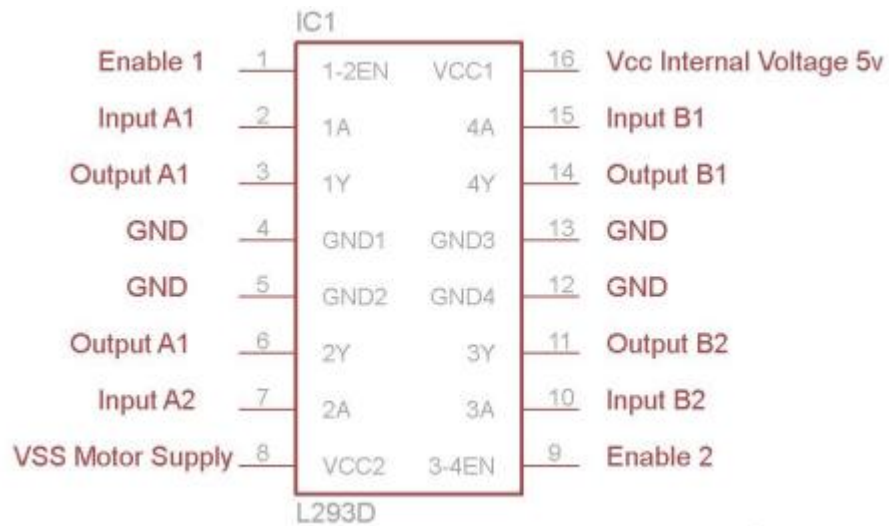


Fig 4.4 Pin diagram of L293D Motor driver

4.3 Resistors and Capacitors

4.3.1 Role of Resistors and Capacitors

1. Coupling Capacitors

- **Function:** Block DC components in the audio signal while allowing AC (DTMF tones) to pass.
- **Placement:** Connected between the audio source and IN+ (Pin 1) of MT8870.

2. Filter Capacitors

- **Function:** Stabilize the power supply by filtering out noise or ripples.
- **Placement:** Connected across VDD and GND near the IC power pins.

3. Oscillator Capacitors

- **Function:** Work with the crystal oscillator to provide a stable clock signal for MT8870.
- **Placement:** Connected to OSC1 (Pin 10) and OSC2 (Pin 11).

4. Resistors for Gain Adjustment

- **Function:** Set the gain of the internal operational amplifier in MT8870.
- **Placement:** Connected to GS (Pin 3).

5. Pull-down Resistors

- **Function:** Ensure binary output pins (D0–D3) remain at a defined state when no key is pressed.
- **Placement:** Connected to D0–D3 and GND.

6. Base Resistors for Transistors

- **Function:** Limit the current to the base of the transistor in the relay driver circuit.
- **Placement:** Connected between the microcontroller and the transistor base.



Fig 4.5 100k ohm resistor



Fig 4.6 1k ohm resistor



Fig 4.7 0.1uF Ceramic Capacitor

4.4 Crystal Oscillator

Key Elements of an Oscillator

1. Amplifier

- Provides gain to maintain oscillations.
- Commonly implemented using an operational amplifier, transistor, or vacuum tube.

2. Feedback Network

- Returns a portion of the output signal back to the input.
- Ensures that the feedback is in-phase (positive feedback) and of the correct magnitude to sustain oscillations.

3. Energy Storage Components

- Capacitors and inductors store energy and determine the frequency of oscillation.
- The resonance condition of these components sets the natural oscillation frequency.

4. Initial Disturbance

- A small noise or disturbance starts the oscillation. The positive feedback amplifies this until steady-state oscillations are achieved.

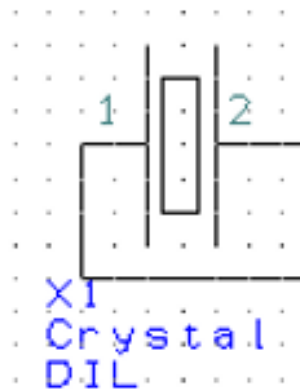


Fig 4.8 Crystal in DesignSpark

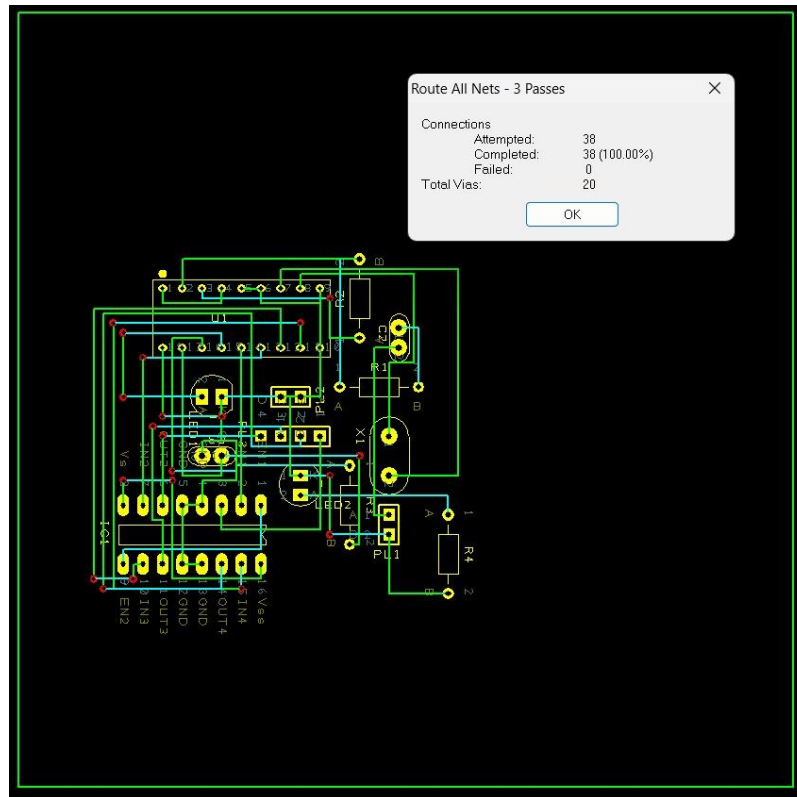


Fig 5.2 PCB Layout Result

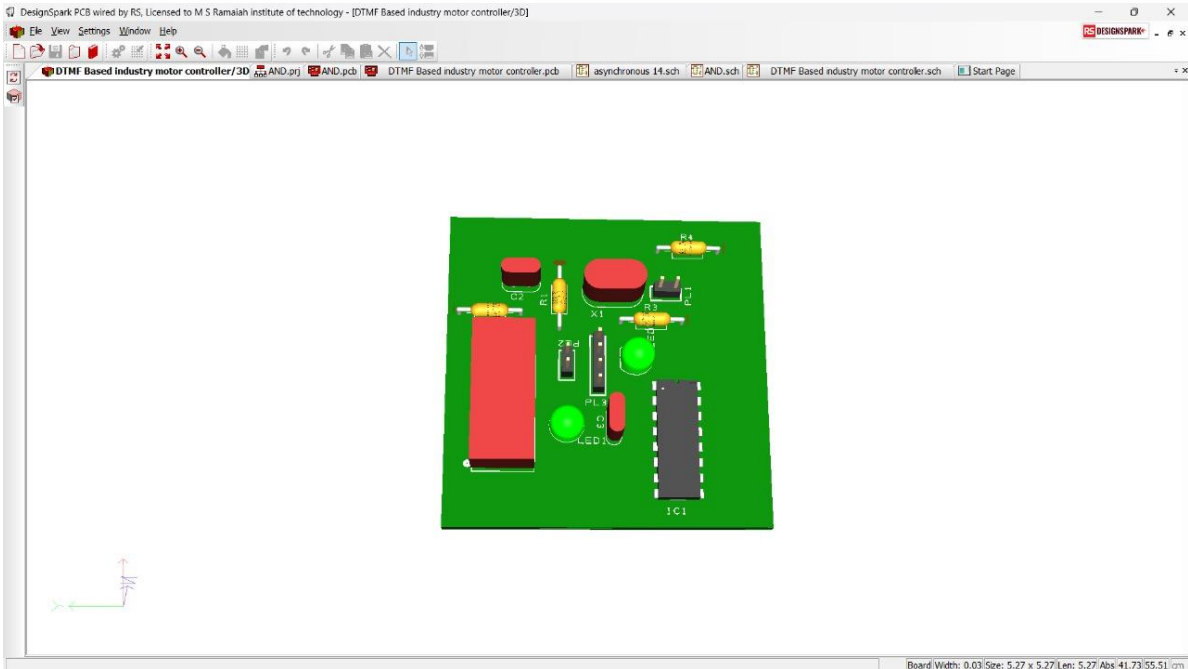


Fig 5.3 3D View of the Chip

Chapter 6

Future Scope

6.1 Enhanced Integration with IoT

- **Internet of Things (IoT) Connectivity:** Future iterations of DTMF-based motor controllers could integrate with IoT platforms, enabling centralized monitoring and control through cloud-based systems. This would allow operators to oversee and manage multiple motors across different locations from a single interface.
 - **Data Analytics:** With IoT integration, data collected from motor operations could be analyzed to predict maintenance needs, optimize performance, and reduce downtime.
 - **Remote Diagnostics:** Advanced diagnostic features can be implemented, enabling technicians to remotely troubleshoot issues without visiting the site physically.
-

6.2 Use in Smart Manufacturing

- **Industry 4.0:** The motor controller can become a vital component of smart manufacturing setups, where automated systems are interconnected. DTMF-based controllers could complement other technologies like sensors and robotics to create efficient, automated production lines.
 - **Flexible Operations:** By enabling remote control of motors, industries can reconfigure production lines quickly and efficiently without significant manual intervention.
-

6.3 Cost-Effective Automation for SMEs

- **Small and Medium Enterprises (SMEs):** Many SMEs face budget constraints when adopting advanced automation technologies. The DTMF-based motor controller provides a low-cost alternative, making automation more accessible to these businesses.
 - **Scalable Implementation:** The system can be scaled as needed, allowing SMEs to start small and expand their automation capabilities over time.
-

6.4 Integration with Renewable Energy Systems

- **Renewable Energy Plants:** In solar or wind energy systems, motors play a crucial role in tracking sun and wind directions. DTMF-based controllers could be used to remotely

manage these motors, ensuring maximum energy efficiency.

- Sustainability Goals: As industries aim to achieve sustainability goals, remote motor control can reduce energy wastage by operating motors only when necessary.
-

6.5 Applications in Hazardous Environments

- Remote Operations in Risky Areas: The DTMF-based system is ideal for industries where motors are located in hazardous environments, such as chemical plants, mining operations, or oil rigs. Remote control reduces the need for human presence in these areas, enhancing safety.
 - Disaster Management: In emergencies, motors in critical locations can be stopped or started remotely, aiding in disaster prevention and control.
-

6.6 Enhanced Security Features

- Encryption of DTMF Signals: To prevent unauthorized access or interference, future systems could implement encrypted DTMF signals, ensuring only authenticated users can control the motors.
 - Access Control: Integration of user authentication methods, such as PIN codes or biometric verification, could enhance security.
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6.7 Transition to Advanced Communication Technologies

- Beyond DTMF: While DTMF is simple and effective, the system could transition to more advanced communication technologies like GSM, Bluetooth, or Wi-Fi for greater versatility.
 - Hybrid Systems: Future designs could incorporate hybrid communication methods, allowing users to switch between DTMF and other technologies depending on network availability.
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6.8 Use in Agricultural Automation

- Irrigation Systems: Farmers could use DTMF-based controllers to remotely operate irrigation pumps, saving water and reducing labor costs.
 - Smart Farming: Integration with soil moisture sensors and weather data could enable fully automated farming solutions.
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6.9 Energy Efficiency and Optimization

- Variable Frequency Drives (VFDs): Future systems could integrate VFDs to control motor speed and torque, optimizing energy usage based on the load requirements.
 - Energy Monitoring: Adding energy monitoring modules would help industries track power consumption and implement energy-saving strategies.
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6.10 Educational and Training Applications

- Hands-On Learning: The simplicity and affordability of DTMF-based controllers make them ideal for educational purposes, enabling students to learn about motor control and automation.
 - Skill Development: Training programs could use these systems to teach workers how to operate and maintain automated industrial equipment.
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6.11 Expansion to Diverse Industrial Sectors

- Logistics and Warehousing: The system can be used to control conveyor belts and automated guided vehicles (AGVs), streamlining material handling processes.
 - Construction: Remote control of motors in cranes, lifts, and other machinery can improve safety and efficiency on construction sites.
 - Healthcare: In hospitals, DTMF-based systems can control motors in medical equipment, ensuring smooth operations with minimal manual intervention.
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6.12 Development of Portable and Compact Systems

- Miniaturization: Future designs could focus on reducing the size and weight of the system, making it more portable and easier to install.
- Plug-and-Play Solutions: Compact designs with plug-and-play functionality could simplify deployment and maintenance.

Chapter 7

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